Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Enterprise Java

Practical – VIII

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Class : TYIT	Date/Time of Submission : 7/08/2026

Aim: Implement the following EJB applications with different types of Beans.

A). Develop a simple EJB application to demonstrate Servlet Hit count using Singleton Session Beans.

CODE:-

index.java

```
<DOCTYPE html>
 <html>
 <head>
 <title>Hit Count Demo</title>
 </head>
 <body>
 <form action="CCClient" method="post">
 <input type="submit"
 value="Take Hit Count demo with EJB Session Singleton Bean"/>
 </form>
 </body>
 </html>
```

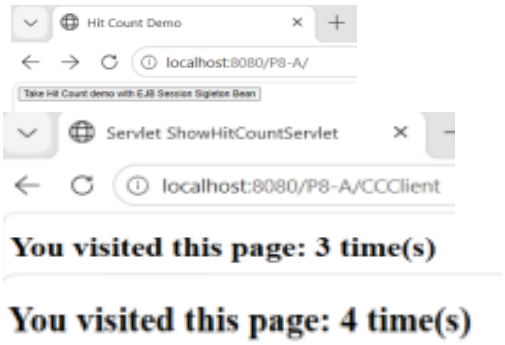
STHitCountBean.java

```
package EJPracts;
import javax.ejb.Singleton;
@Singleton
public class STHitCountBean {
 private int count;
 public synchronized int incrHitCount() {  return count++;
 }
}
```

CCClient.java

```
package servlet;
import javax.servlet.annotation.WebServlet; import
EJPracts.STHitCountBean;
import java.io.IOException;
import java.io.PrintWriter;
import javax.ejb.EJB;
import javax.servlet.ServletException; import
javax.servlet.http.HttpServlet; import
javax.servlet.http.HttpServletRequest; import
javax.servlet.http.HttpServletResponse;
@WebServlet(urlPatterns = {"/CCClient"}) public class
CCClient extends HttpServlet {
 @EJB
 STHitCountBean cntBean;
 protected void processRequest(HttpServletRequest  request,
HttpServletResponse response)
 throws ServletException, IOException {
response.setContentType("text/html;charset=UTF-8");  try
(PrintWriter out = response.getWriter()) {
out.println("<!DOCTYPE html>");
 out.println("<html>");
 out.println("<head>");
 out.println("<title>Servlet
ShowHitCountServlet</title>");
 out.println("</head>");
 out.println("<body>");
 out.println("<h1>You visited this page: " +
cntBean.incrHitCount() + " time(s)</h1>");
 out.println("</body>");
 out.println("</html>");
 }
 }
}
```

OUTPUT:-



1

B). Develop a JSP application to demonstrate Form Processing and Input Validation.

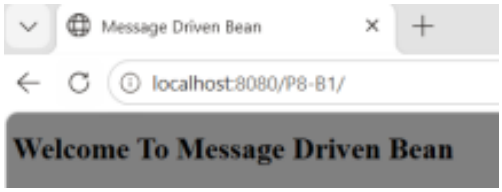
CODE:-

index.jsp

```
 <%@page import="javax.jms.JMSException"%>
<%@page import="javax.jms.TextMessage"%>
<%@page import="javax.jms.MessageProducer"%>
<%@page import="javax.jms.Session"%>
```

OUTPUT:-

```
<%@page import="javax.jms.Connection"%>
<%@page import="javax.jms.Queue"%>
<%@page import="javax.jms.ConnectionFactory"%>
<%@page import="javax.naming.InitialContext"%>
<%@page import="javax.naming.NamingException"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
 <meta http-equiv="Content-Type"
 content="text/html; charset=UTF-8">
 <title>Message Driven Bean</title>
</head>
<body style="background-color:gray;">
<h1>Welcome To Message Driven Bean</h1>
<%
 Connection conn = null;
 Session mysess = null;
 try {
 InitialContext ctx = new InitialContext();
Queue queue =
 (Queue) ctx.lookup("jms/Queue");
ConnectionFactory connFact =
 (ConnectionFactory)
 ctx.lookup("jms/QueueFactory");  conn
= connFact.createConnection();  mysess
= conn.createSession(
 false,
 Session.AUTO_ACKNOWLEDGE  );
 MessageProducer msgProducer =
 mysess.createProducer(queue);
 TextMessage msg =
 mysess.createTextMessage();
 msg.setText(request.getRemoteAddr());
msgProducer.send(msg);
 } catch (NamingException ne) {
 System.out.println(
 "Naming Exception occurred: " + ne.toString()  );
 } catch (JMSException e) {
 System.out.println(
 "Exception occurred: " + e.toString()  );
 } finally {
```




```
mysql> select * from visitorstat;
+---------------------+-----------------+------------+
| #visitdate          | host            | visitcount |
+---------------------+-----------------+------------+
| 2026-08-29 13:45:56 | 0:0:0:0:0:0:0:1 |          4 |
+---------------------+-----------------+------------+
1 row in set (0.00 sec)
```

```
 if (mysess != null)
 try {
 mysess.close();
 } catch (JMSException ignore) {}  if
(conn != null)
 try {
 conn.close();
 } catch (JMSException ignore) {}
 }
 %>
 </body>
 </html>
```

VStatelessBean.java

```
package EJPracts;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
import java.sql.Statement;
import javax.ejb.Stateless;
@Stateless
public class VStatelessBean {
 private Connection getConnection() throws
SQLException, ClassNotFoundException {  // Try
MySQL 8+ driver first, fall back to legacy  driver
 try {
 Class.forName("com.mysql.jdbc.Driver");  }
catch (ClassNotFoundException e) {
Class.forName("com.mysql.jdbc.Driver")
 }
 // Update password if your MySQL root account has  one
(e.g. "root" or "")
 String connStr =
"jdbc:mysql://localhost:3306/reg?useSSL=false&allowPub
l icKeyRetrieval=true";
 return DriverManager.getConnection(connStr, "root",
"zeus");
 }
 public void addVisitor(String host) {
 Connection conn = null;
 Statement stmt = null;
 try {
 conn = getConnection();
 stmt = conn.createStatement();
 // Try to insert first
 String query = "INSERT INTO visitorstat (host,
visitcount) VALUES('" + host + "', 1)";
 stmt.executeUpdate(query);
 System.out.println("Inserted visitor: " + host);   }
catch (SQLException e) {
 // If host already exists (duplicate key), update
visitcount
 try {
 if (conn != null) {
 stmt = conn.createStatement();
 String updateQuery = "UPDATE visitorstat  SET
visitcount = visitcount + 1 WHERE host = '" + host +  "'";
 stmt.executeUpdate(updateQuery);
System.out.println("Updated visitor count for:  " + host);
 }
 } catch (SQLException ex) {
 System.err.println("Cannot update visitorstat: " +
ex.getMessage());
 }
 } catch (Exception ex) {
 System.err.println("Database Connection Failed: " +
ex.getMessage());
 } finally {
```

```
 if (stmt != null) try { stmt.close(); } catch
(SQLException ignore) {}
        if (conn != null) try { conn.close(); } catch
       (SQLException ignore) {}
        }
        }
        }
```

MsgDrvnBean.java

```
package EJPracts;
import javax.annotation.Resource;
import javax.ejb.ActivationConfigProperty;
import javax.ejb.EJB;
import javax.ejb.MessageDriven;
import javax.ejb.MessageDrivenContext;
import javax.jms.JMSException;
import javax.jms.Message;
import javax.jms.MessageListener;
import javax.jms.TextMessage;
@MessageDriven(activationConfig = {
 @ActivationConfigProperty(propertyName =
"destinationLookup", propertyValue = "jms/Queue"),
@ActivationConfigProperty(propertyName =
"destinationType", propertyValue =
"javax.jms.Queue") })
public class MsgDrivnBean implements MessageListener
{  @EJB
 private VStatelessBean vs;
 @Resource
 private MessageDrivenContext mdc;
 public MsgDrivnBean() {
 }
 @Override
 public void onMessage(Message message) {
try {
 if (message instanceof TextMessage) {  TextMessage
msg = (TextMessage) message;
vs.addVisitor(msg.getText());
 }
 } catch (JMSException e) {
 mdc.setRollbackOnly();
 }
 }
 }
```

C). Develop a simple visitor Statistics application using Message Driven Bean [MDB].

CODE:-

index.jsp

```
<%@page import="javax.naming.InitialContext"%>
<%@page import="EJPracts.MarksEntryBean"%>
<%@page contentType="text/html" pageEncoding="UTF
8"%>
<!DOCTYPE html>
<%
 String message = "";
 if (request.getParameter("insertMarks") != null) {
String name = request.getParameter("stuName");
String marks1 = request.getParameter("sub1");
String marks2 = request.getParameter("sub2");
String marks3 = request.getParameter("sub3");
MarksEntryBean marksEntry = null;

 // Try JNDI lookups first
 try {
 InitialContext ic = new InitialContext();  try {
 marksEntry = (MarksEntryBean)
ic.lookup("java:global/P8-
C/MarksEntryBean!EJPracts.MarksEntryBean")
;  } catch (Exception e1) {
 try {
 marksEntry = (MarksEntryBean)
ic.lookup("java:global/P8-C/MarksEntryBean");
} catch (Exception e2) {
 marksEntry = (MarksEntryBean)
ic.lookup("java:module/MarksEntryBean");
}
 }
 } catch (Exception e) {
 //  fallback  to  direct  object  instantiation
marksEntry = new MarksEntryBean();  }
 if (marksEntry == null) {
 marksEntry = new MarksEntryBean();  }
 try {
 marksEntry.addMarks(name, marks1, marks2,
marks3);
 message = "<p style='color:green; text
align:center;'>Record Inserted Successfully!</p>";
} catch (Exception ex) {
 message = "<p style='color:red; text
align:center;'>Database Error: " + ex.getMessage() +
"</p>";
 }
 }
%>
<html>
<head>
 <meta http-equiv="Content-Type" content="text/html;
charset=UTF-8">
 <title>Marks Entry</title>
</head>
<body style="font-family: Arial; padding: 20px;">
<h2 style="text-align: center;">Enter Details</h2>
<%= message %>
 <form name="result" method="post"
action="index.jsp">
 <table align="center" border="1" cellpadding="8">
<tr>
 <td>Enter Student Name</td>
 <td><input type="text" name="stuName"  required
/></td>
 </tr>
 <tr>
        <td>Enter Subject 1 Marks</td>
        <td><input type="text" name="sub1" required  /></td>
        </tr>
        <tr>
        <td>Enter Subject 2 Marks</td>
        <td><input type="text" name="sub2" required  /></td>
        </tr>
        <tr>
        <td>Enter Subject 3 Marks</td>
```

```
<td><input type="text" name="sub3" required /></td>
</tr>
<tr>
<td colspan="2" align="center">
<input type="submit" name="insertMarks"
value="Insert Marks" />
</td>
</tr>
</table>
</form>
</body>
</html>
```

MarksEntryBean.java

```
package EJPracts;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
import java.sql.Statement;
import javax.ejb.Stateful;
@Stateful
public class MarksEntryBean {

String query = "INSERT INTO marks(stuName, sub1,
sub2, sub3) VALUES('"
+ stuname + "', '" + marks1 + "', '" + marks2 + "', '" +
marks3 + "')";
stmt.executeUpdate(query);
System.out.println("Marks inserted successfully for: " +
stuname);
} catch (Exception e) {
System.err.println("Failed to insert: " +
e.getMessage());
private Connection getConnection() throws Exception {
try {
Class.forName("com.mysql.cj.jdbc.Driver"); }
catch (ClassNotFoundException e) {
Class.forName("com.mysql.jdbc.Driver"); }
// Change root password if you have set one
String connStr =
"jdbc:mysql://localhost:3306/reg?useSSL=false&allowPub
l icKeyRetrieval=true";
return DriverManager.getConnection(connStr, "root",
"zeus");
}
public void addMarks(String stuname, String marks1,
String marks2, String marks3) {
Connection conn = null;
Statement stmt = null;
try {
conn = getConnection();
stmt = conn.createStatement(

} finally {
if (stmt != null) try { stmt.close(); } catch
(SQLException ignore) {}
if (conn != null) try { conn.close(); } catch
(SQLException ignore) {}
}
}
}
```

OUTPUT:-

